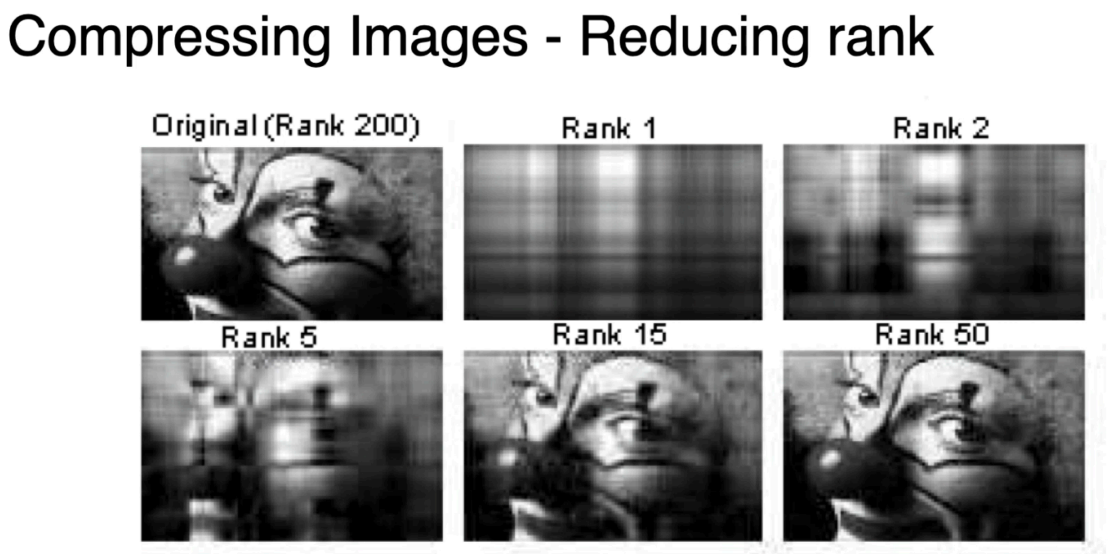
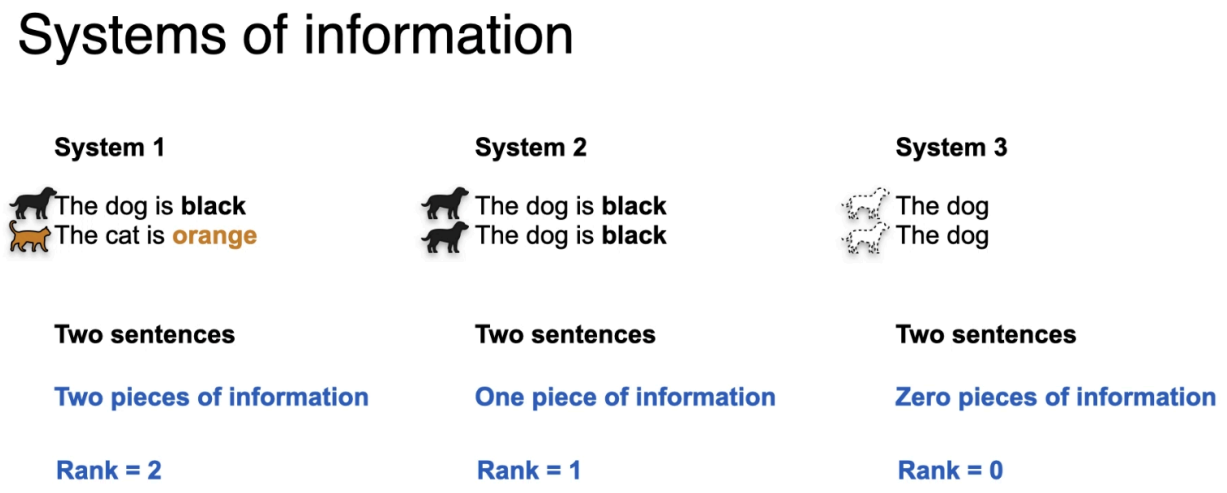
Rank of a matrix

The rank of a matrix measures how much information that matrix or its corresponding system of linear equations carries
It represents the number of non-zero rows or columns of a matrix. In other words, the number of linearly independent rows or columns in a matrix is called the rank of a matrix

SVD = Singular Value Decomposition, We use it to reduce the rank of the original photo so that it does not require a lot of storage or computation power.





Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Rank = 2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

1	1
2	2

Rank = 1

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

0	0
0	0

Rank = 0

Two equations

Zero pieces of information

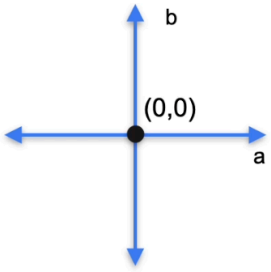
Rank = 0

Rank and solutions to the system

1	1
1	2

Rank = 2

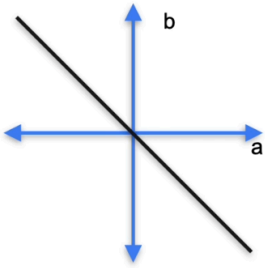
Dimension of solution space = 0



1	1
2	2

Rank = 1

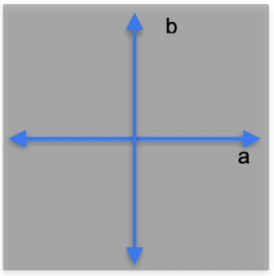
Dimension of solution space = 1



0	0
0	0

Rank = 0

Dimension of solution space = 2



Rank of a matrix

1	1
1	2

Rank = 2

Dimension of solution space = 0

1	1
2	2

Rank = 1

Dimension of solution space = 1

0	0
0	0

Rank = 0

Dimension of solution space = 2

Rank = 2 - (Dimension of solution space)

Rank = Number of rows that is complete, not redundant

Solutions: Rank of a matrix

Determine the rank of the following two matrices

Matrix 1: Since the solution space had dimension 0, the rank is **2**.

5	1
-1	3

Matrix 1: Since the solution space had dimension 1, the rank is **1**.

2	-1
-6	3

Row echelon form, singularity, and rank

5	1
4	-3

→

1	0.2
0	1

Rank 2
2 ones in the diagonal

5	1
10	2

→

1	0.2
0	0

Rank 1
1 one in the diagonal

0	0
0	0

→

0	0
0	0

Rank 0
0 ones in the diagonal

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Rank = 3

Matrix 2

1	1	1
1	1	2
1	1	3

Rank = 2

Matrix 3

1	1	1
2	2	2
3	3	3

Rank = 1

Matrix 4

0	0	0
0	0	0
0	0	0

Rank = 0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

Reduced row echelon form

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

Rank 5

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Rank 3

- Is in row echelon form
- Each pivot is a 1
- Any number above a pivot is 0
- Rank of the matrix is the number of pivots

Reduced row echelon form

Row echelon form

3	*	*	*	*
0	0	2	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

1	*	*	*	*
0	0	1	*	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Divide each row by the value of the pivot

Reduced row echelon form

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Turn anything above a pivot to 0

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Subtract 2 times the second row from the first one

1	0	-5
0	1	4
0	0	1

Add 5 times the third row to the first one

1	0	0
0	1	4
0	0	1

Subtract 4 times the third row from the second one

Reduced row echelon form

1	0	0
0	1	0
0	0	1